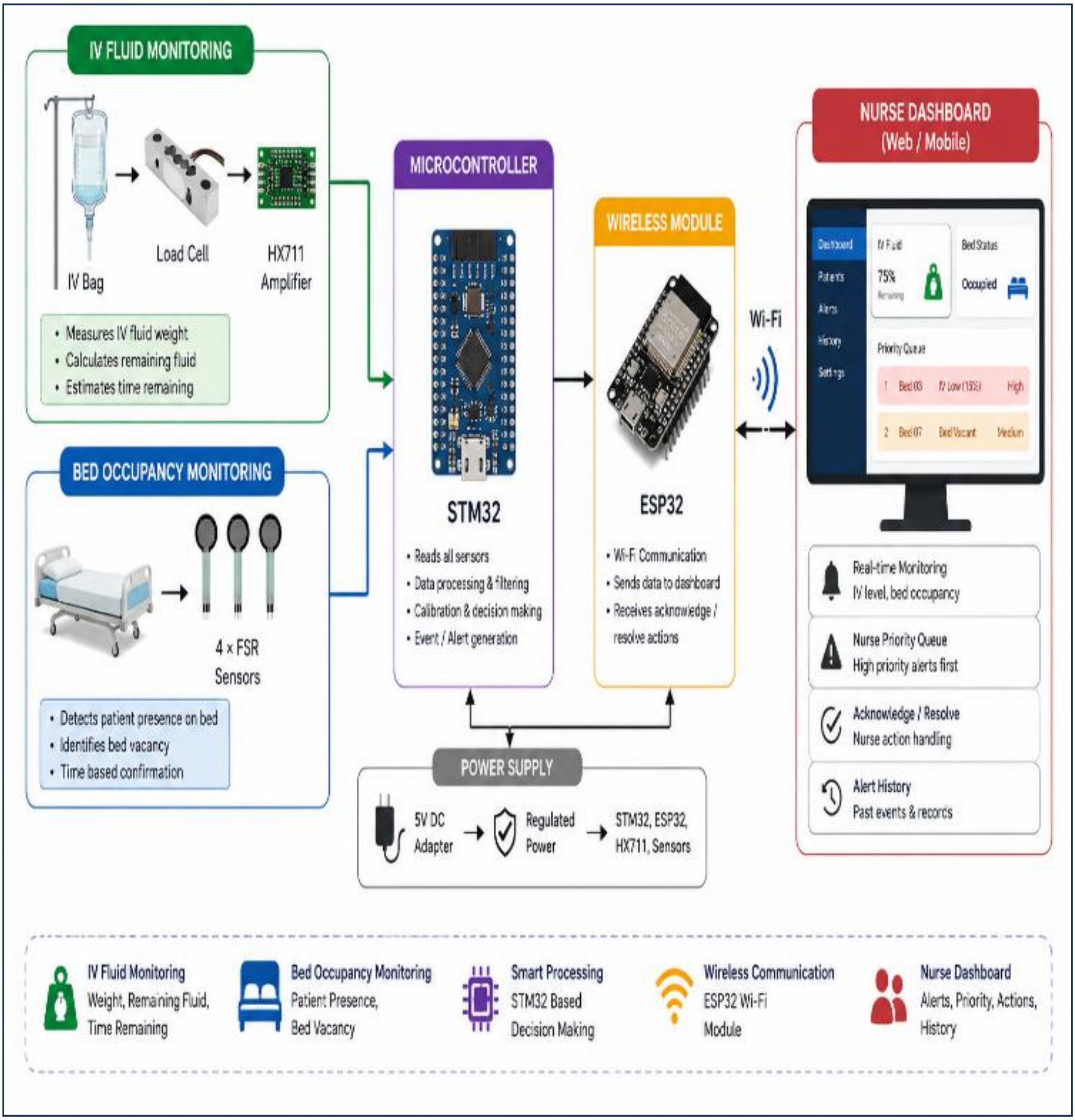
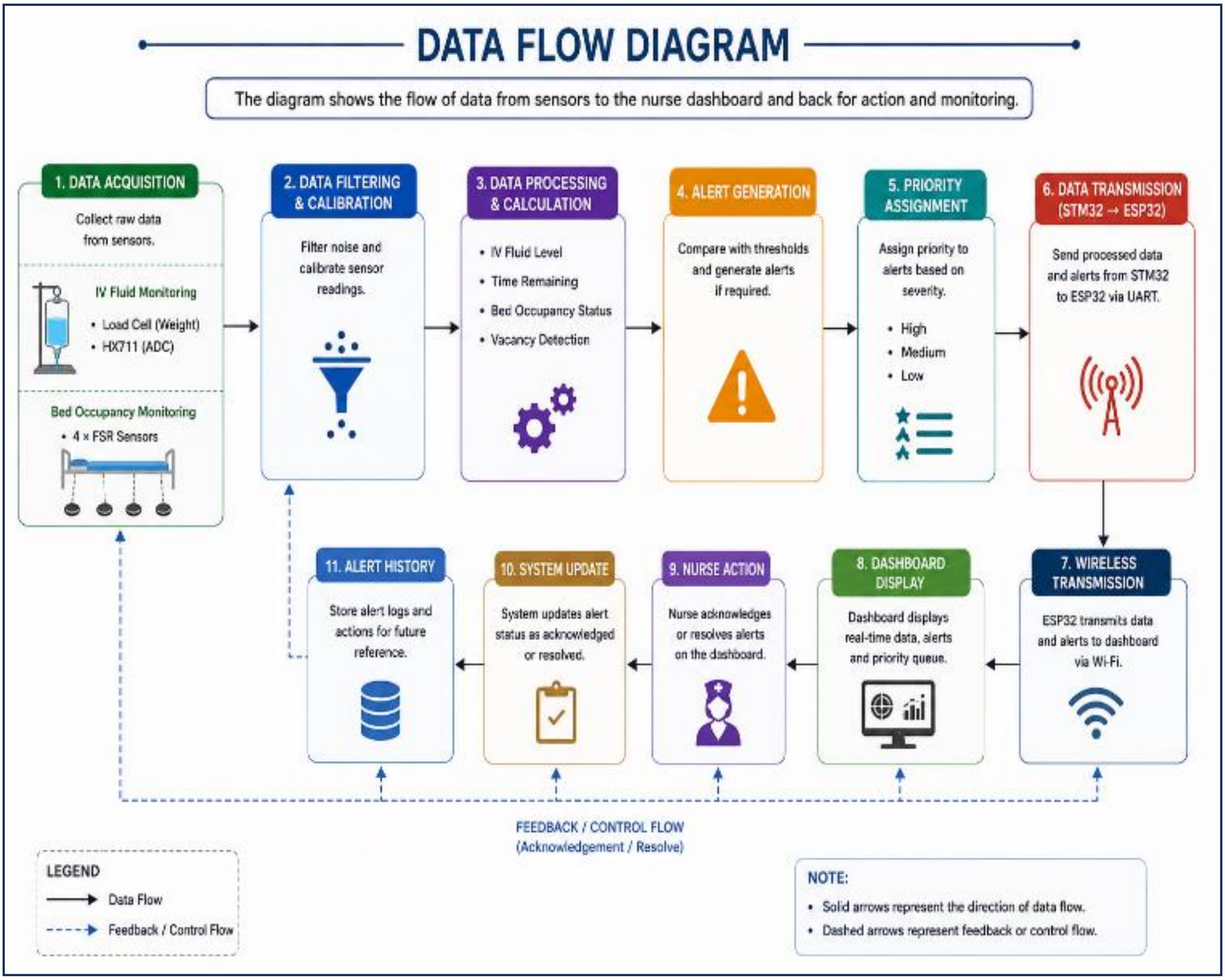
SYSTEM ARCHITECTURE:

Overall Solution Architecture: Shows the major components of the proposed system and their interaction from sensing to nurse notification.

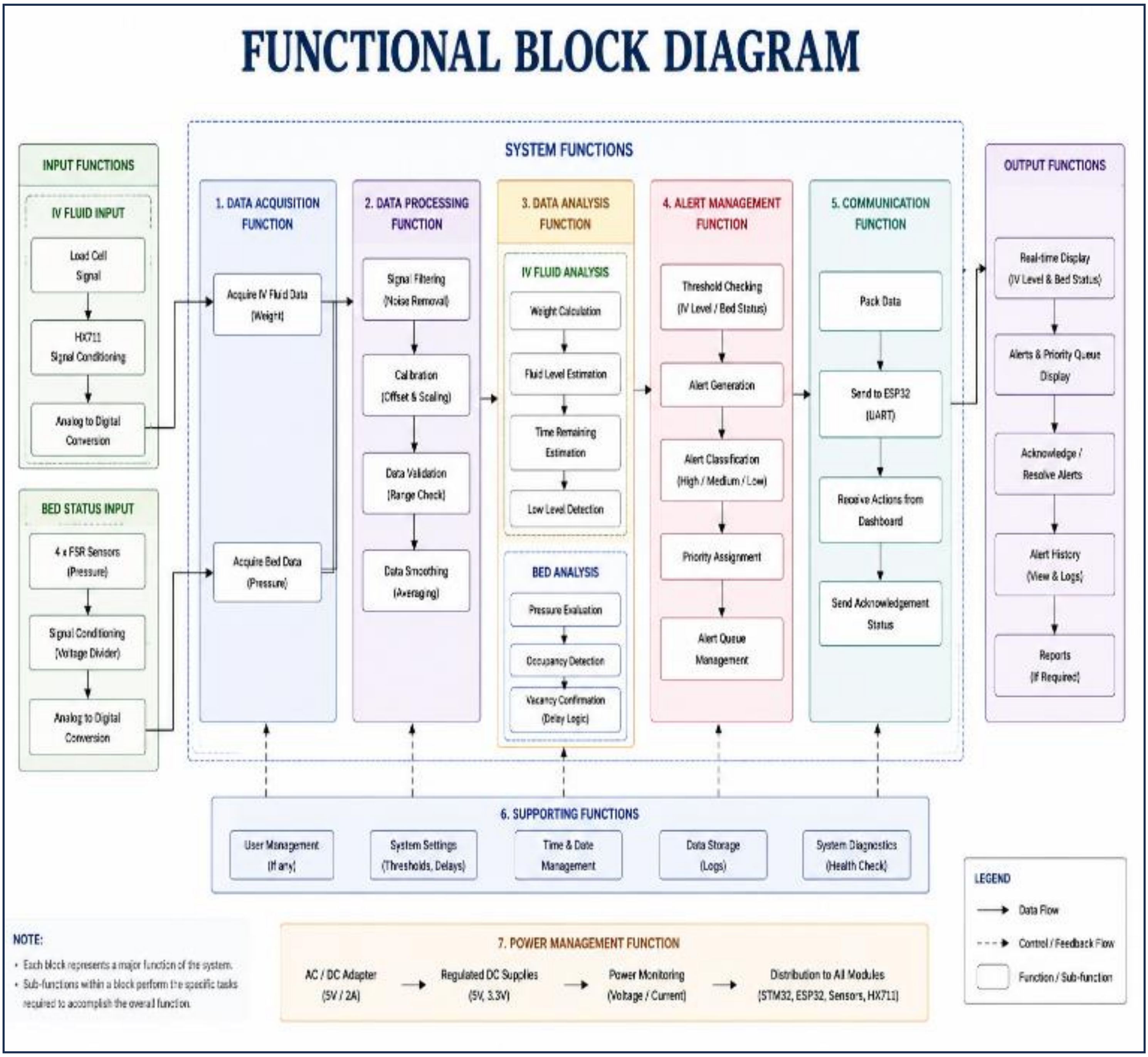


Data Flow Diagram: Shows how sensor data moves from the sensing units through the STM32 and ESP32 to the nurse dashboard.



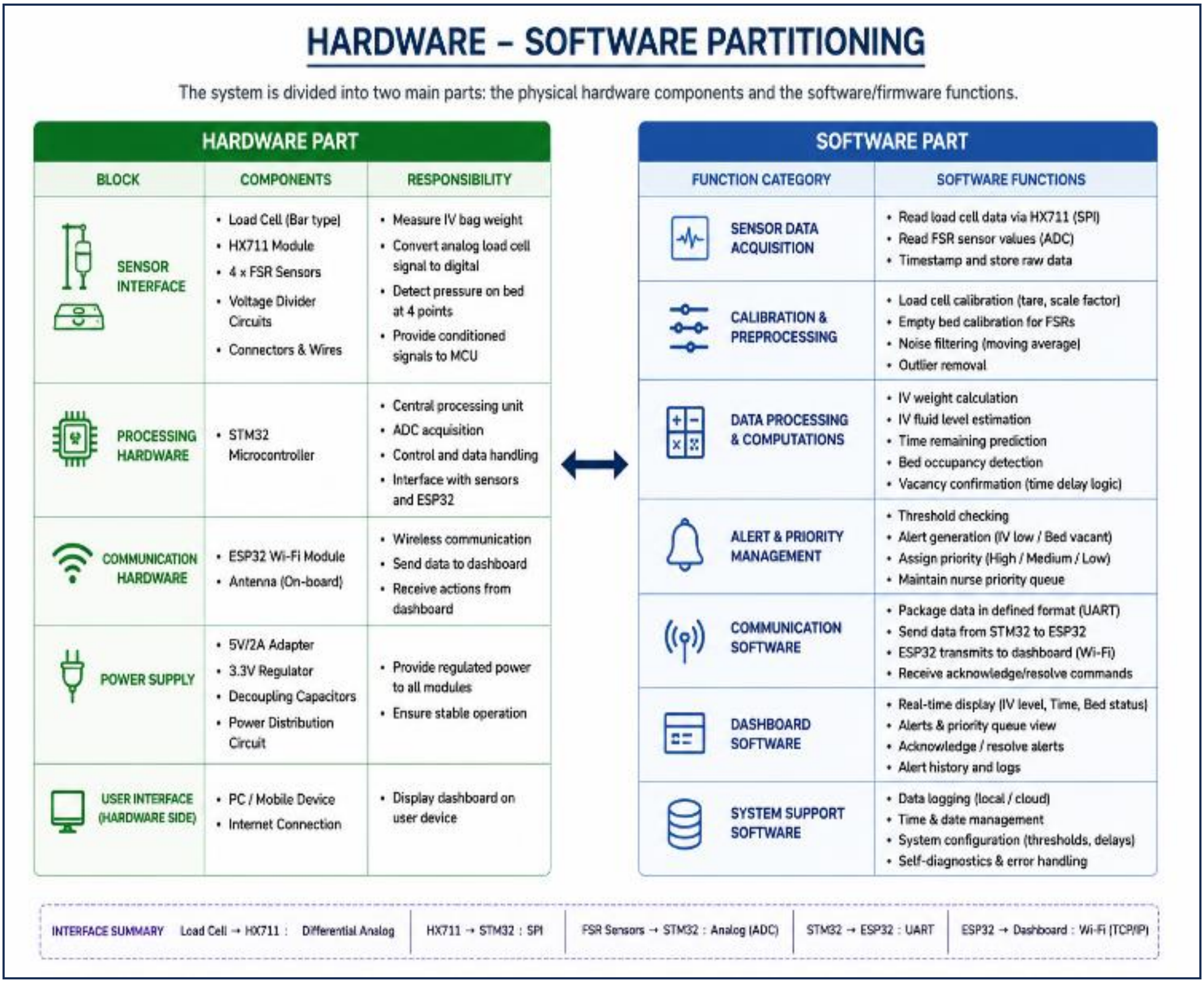
Functional Block Diagram:

Shows the functional operations performed by each hardware/software block and their connections.



Hardware–Software Partitioning:

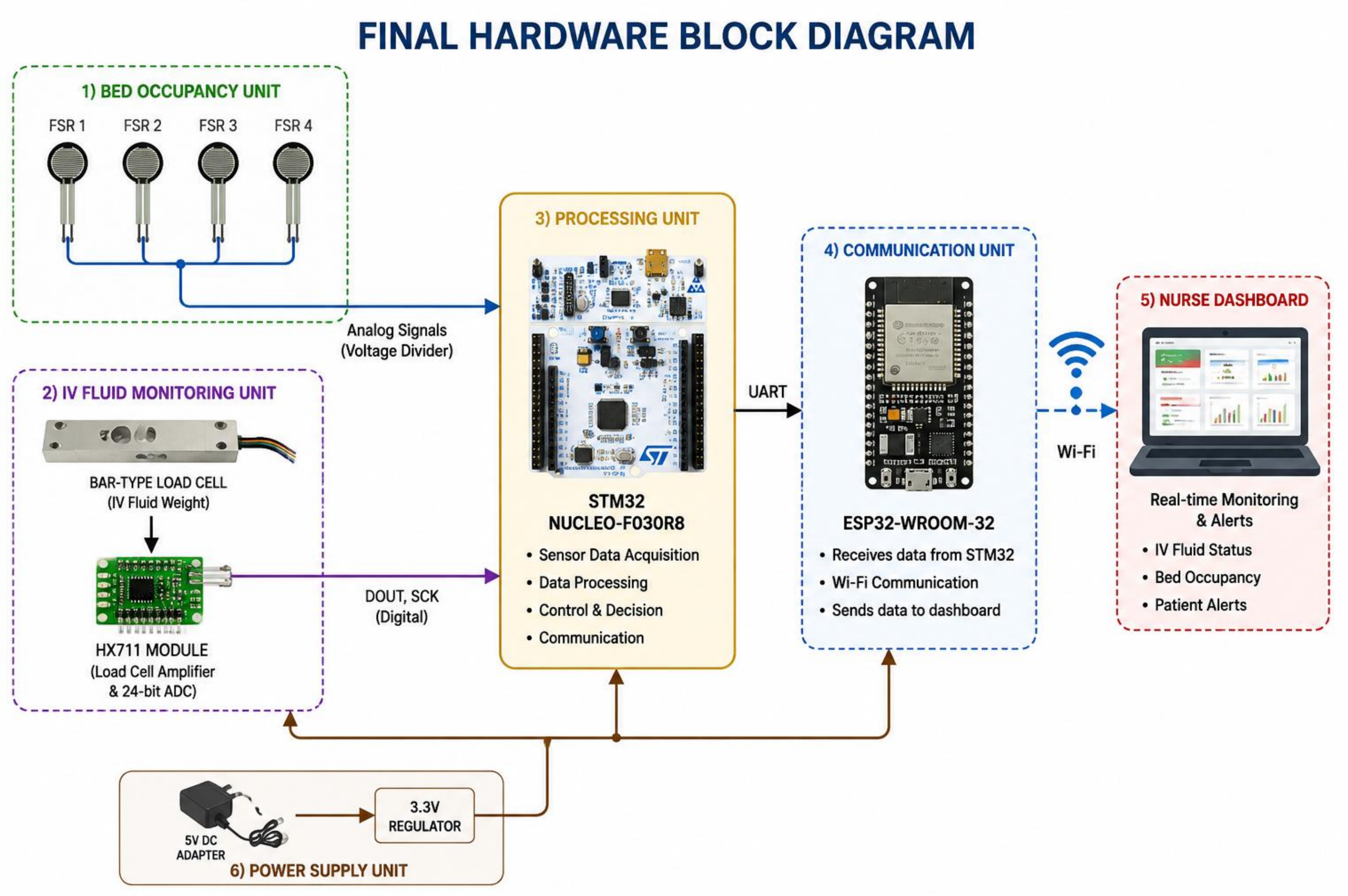
Shows which functions are implemented using hardware components and which are handled by software.



HARDWARE DESIGN:

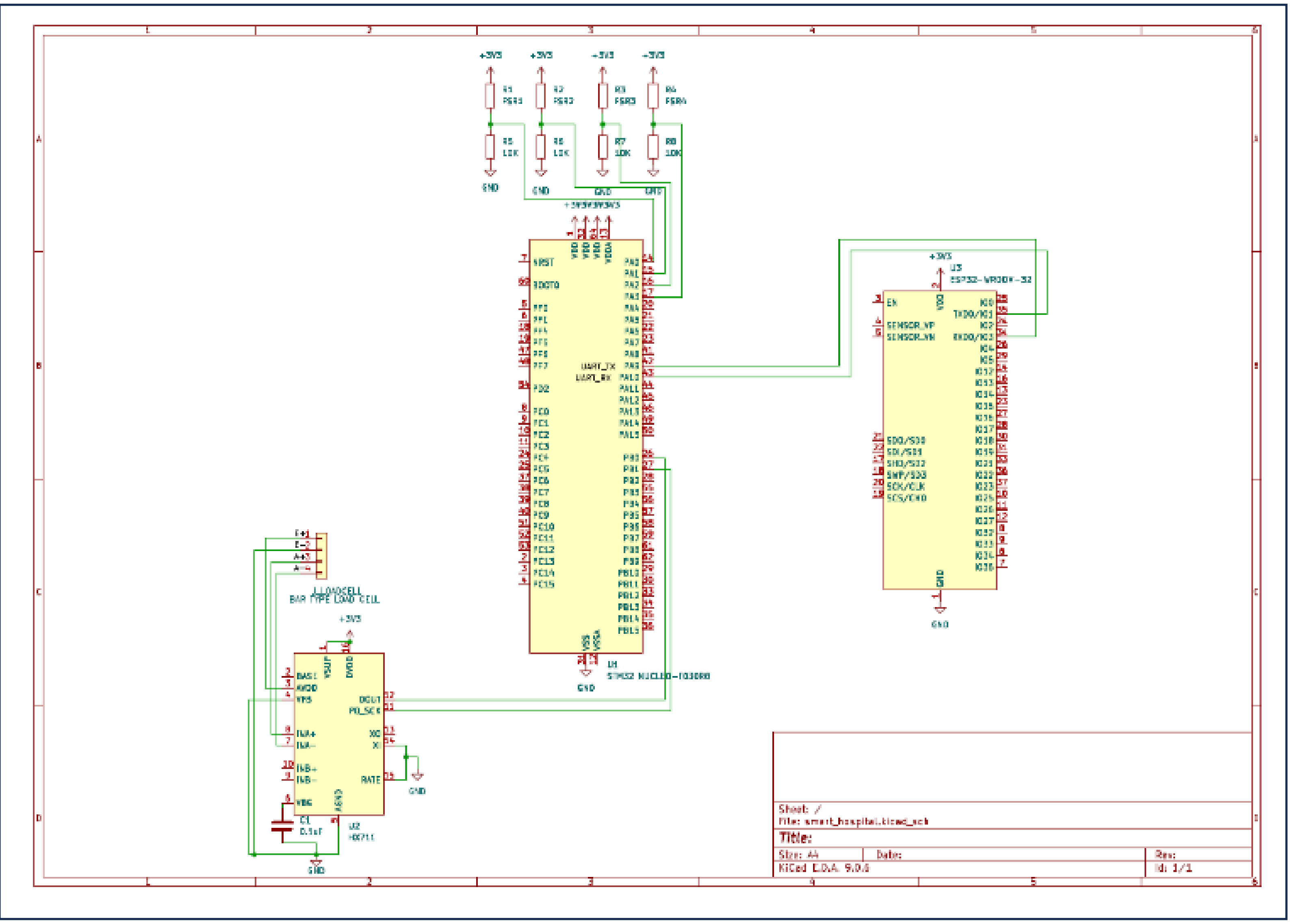
FINAL BLOCK DIAGRAM:

The block diagram shows the major hardware components and their interconnections for bed occupancy and IV fluid monitoring.



CIRCUIT SCHEMATICS:

The schematic shows the electrical connections between the FSR sensors, load cell, HX711, STM32 Nucleo, and ESP32.



COMPONENT SELECTION:

Component	Why needed
STM32 NUCLEO-F030R8	Processes sensor readings
4 FSR Sensors	Detects patient presence on the bed
Bar-type Load Cell	Measures IV fluid weight
HX711	Amplifies and converts load-cell signals
ESP32-WROOM-32	Sends data through Wi-Fi
10 kΩ Resistors	Forms voltage dividers with FSRs
3.3 V Power Supply	Powers the electronic components
0.1 µF Capacitors	Reduces power-supply noise

BILL OF MATERIALS:

S.No.	Component	Specification	Quantity
1	STM32 NUCLEO-F030R8	STM32F030R8T6, 32-bit MCU	1
2	FSR Sensor	Force Sensitive Resistor	4
3	Bar-type Load Cell	Weight measurement	1
4	HX711 Module	24-bit ADC / Load-cell amplifier	1
5	ESP32-WROOM-32	Wi-Fi module	1
6	Resistor	10 kΩ	4
7	Capacitor	0.1 µF	As required
8	Connecting Wires	Jumper wires	As required
9	Breadboard/PCB	Prototype assembly	1